and harmful **pathogenic** bacteria which are diversely populated. Cyanobacteria are photosynthetic blue green algae which produce oxygen. These had played a key role in the changes of atmospheric oxygen levels from anaerobic to aerobic during the early geologic periods.

3. Domain Eukarya (Eukaryotes)

Eukaryotes are animals which have true nucleus and membrane bound organelles. DNA in the nucleus is arranged as a linear chromosome with histone proteins, ribsosomes of 80S type in the cytosol and 70S type in the chloroplast and mitochondria. Organisms in this domain are classified under kingdoms, namely, Protista, Fungi, Plantae and Animalia.

In 1987, **Cavalier-Smith** revised the six kingdom system to **Seven Kingdom system.** The concept of super kingdom was introduced and revised to seven kingdom classification. The classification is divided into two Super Kingdoms (Prokaryota and Eukaryota) and seven kingdoms, two Prokaryotic Kingdoms (Eubacteria and Archaebacteria) and five Eukaryotic Kingdoms (Protozoa, Chromista, Fungi, Plantae and Animalia) (Table 1.1).

Curd is one of the best sources of probiotics, which are friendly bacteria that can improve our health. e.g. *Lactobacillus sp.*

The Three-Domain System

Archaea | Bacteria | Eukarya

The traditional Five – Kingdom system

Monera | Protista | Fungi | Plantae | Animalia

The Six –Kingdom system

Bacteria | Archaea | Protista | Fungi | Plantae | Animalia

The Seven – Kingdom system

Eubacteria | Archaea bacteria | Protozoa | Chromista | Fungi | Plantae | Animalia

Table 1.1 Five Kingdom Classification

Salient features	KINDS OF KINGDOM				
	Monera	Protista	Fungi	Plantae	Animalia
Cell type	Prokaryotic	Eukaryotic	Eukaryotic	Eukaryotic	Eukaryotic
Cell wall	Non-cellulose structure	Present in some	Present	Present	Absent
Body organisation	Cellular	Cellular	Multicelluar Tissue	Tissue Organ	Tissue Organ Organ system
Mode of nutrition	Autotrophic Heterotrophic	Autotrophic Heterotrophic	Heterotrophic	Autotrophic	Heterotrophic

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